

Devi Ruwali, Data Engineer | DataOps

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Objective

Seeking a **Senior Data Engineer** role in a forward-thinking organization where **creativity, sincerity, skill, and performance** are the measures of growth and recognition, backed by **6+ years** of expertise in **cloud-native data pipelines, distributed architectures, and DevOps best practices** to deliver impactful, **data-driven** business outcomes.

Summary

- Designed **cloud-native data pipelines** in AWS and GCP, leveraging **Terraform** for infrastructure as code and enabling automated, scalable deployments across multiple environments.
- Engineered **Big Data ETL** workflows using **Spark (PySpark/Scala)** and **Hive**, efficiently processing multi-terabyte data sets and cutting query times from hours to minutes.
- Employed **containerization** with **Docker** and **Kubernetes** to create fault-tolerant microservices, minimizing overhead and simplifying the deployment process in dev/test/prod.
- Built DataOps workflows in **Airflow** and **AWS Step Functions**, reducing **data latency by 50%** and accelerating near real-time analytics for high-impact use cases.
- Automated **CI/CD pipelines** with **Jenkins, GitHub Actions**, and **Git**, implementing comprehensive testing and validation to streamline release cycles.
- Enforced **data governance** via **Collibra** and **Great Expectations**, maintaining lineage, metadata, and compliance standards in regulated environments.
- Secured enterprise data with **IAM roles/policies, encryption at rest/in transit**, and **OAuth2**, ensuring robust **RBAC** to meet strict compliance and prevent unauthorized access.
- Deployed and optimized **Snowflake** data warehouses, integrating advanced partitioning and clustering strategies to halve query durations and maximize performance.
- Integrated **Kafka** and **RabbitMQ** for real-time streaming and asynchronous message flows, boosting operational responsiveness and facilitating rapid event-driven processing.
- Aligned with **Agile/Scrum** practices, collaborating closely with cross-functional teams (Data Scientists, Risk, Analytics) to deliver solutions that meet evolving business needs.
- Utilized multiple languages—**Python, Scala, Java, SQL**, and **JavaScript**—to develop full-stack solutions, from backend processing to frontend integrations.
- Guided **junior engineers** on **DevSecOps** best practices, code optimization, and data engineering frameworks, cultivating a dynamic culture of continuous improvement.
- Developed self-service infrastructure through **Terraform**, cutting manual configuration by 80% and empowering teams to provision their own resources.
- Improved high-volume **Spark** clusters using partitioning, caching, and tuning, enabling consistent handling of large-scale data sets with near real-time processing speeds.
- Created robust **testing** strategies (Postman, Rest Assured, TestNG, and Cucumber) for RESTful APIs, microservices, and SQL validations to ensure data integrity and code reliability.
- Provided **ML-ready datasets** and **real-time dashboards** (e.g., Power BI), driving data-driven decisions for varied stakeholders and accelerating analytics initiatives.
- Embraced **serverless** compute (AWS **Lambda**) alongside **Docker/Kubernetes** to optimize costs, scale processing on demand, and support hybrid cloud environments.

Technical Skills

Programming Languages	Python, Scala, SQL, Java, JavaScript
Data & Analytics	Spark (PySpark, Scala), Hive, Kafka, RabbitMQ, dbt, ML (Forecasting, Analytics), Airflow, AWS Step Functions, Great Expectations, Power BI
Cloud & Infrastructure	AWS (S3, Redshift, EMR, Lambda), GCP, Terraform, Docker, Kubernetes
Databases & Warehouses	Oracle, DB2, Snowflake, Redshift
DevOps & CI/CD	Jenkins, GitHub Actions, Git, CI/CD pipelines, Maven, Microservices Architecture
Security & Governance	IAM roles/policies, Encryption (at rest/in transit), OAuth2, Role-Based Access Control (RBAC), Collibra (Data Governance), Data Lineage Tracking
Testing & Automation	Postman, Rest Assured, TestNG, Cucumber, JUnit, Page Object Model (POM), Automated Scripts, Root-cause Analysis

Frameworks	Java Spring (backend), Flask/FastAPI (Python), React (front-end), Containerized Services (Docker, Kubernetes), Serverless (AWS Lambda)
Project Methodologies	Agile (Scrum), DevSecOps, Continuous Improvement, Sprint Planning, Retrospectives

Professional Experience

Vanguard, Malvern, PA, April 2023 – Present

Data Engineer, Specialist

Portfolio Analytics Tool is a real-time, cloud-native platform for on-demand risk/return diagnostics, portfolio comparisons, and hypothetical performance modeling. It integrates **streaming pipelines (Kafka, Spark)**, **ML-driven forecasting**, and a **self-service portal** for scenario analyses—all fortified by IAM encryption and Collibra-based governance to ensure compliance.

Responsibilities:

- Collaborated with cross-functional stakeholders (**Product, Risk, Analytics**) to translate complex business specifications into **technical designs** and **production-ready code**, delivering on-demand risk analyses and **reducing turnaround times by 30%** for ad-hoc reports.
- Developed and deployed a **highly scalable project in AWS (S3, Redshift, EMR, Lambda)** using **Terraform**, enabling self-service infrastructure deployments and reducing **manual configuration by 80%**.
- Automated ingestion from multiple sources (**Oracle, DB2, Kafka, REST APIs**) and orchestrated **ETL workflows (Airflow)**, **reducing data latency by 50%** and enabling near real-time availability for downstream analytics.
- Leverage containerization (**Docker, Kubernetes**) and **serverless compute** to minimize costs, auto-scale resources, and streamline deployment across dev/test/prod environments.
- Implement **CI/CD pipelines (Jenkins, GitHub Actions)** to build, test, and deploy data transformations (**Python, PySpark, Scala, SQL, dbt**) with comprehensive **code reviews** and branching strategies.
- Optimize **Spark (PySpark/Scala)** and **Hive** jobs for high-volume processing, employing **partitioning, caching, and cluster tuning** for performance gains and **reduced query times**.
- Enforce data governance via **Collibra, Great Expectations, and lineage tracking**, ensuring data quality, **metadata management, and compliance** with enterprise standards.
- Employ robust security (**IAM** roles/policies, **encryption** at rest/in transit) and **role-based access** controls to safeguard data across **AWS** environments and pipelines.
- Monitor pipelines with tools like **CloudWatch, Datadog, and Splunk**, setting **proactive alerts** and conducting **root-cause analyses** for failures or data anomalies.
- Collaborate in **agile squads**, partnering with **Data Scientists, DevOps, and Analysts** to deliver **real-time dashboards** and **ML-ready datasets**.
- Lead all phases of **solution development**, including thorough **testing** for accuracy, explain **technical considerations** to stakeholders, review final deliverables with clients, and provide **data analysis guidance** as needed.
- Mentor **junior engineers** in **data engineering best practices**, **DevSecOps** principles, and **code optimization**, fostering a culture of **continuous improvement**.

Environment: AWS (S3, Redshift, EMR, Lambda), Terraform, Oracle, DB2, Kafka, Airflow, AWS Step Functions, Docker, Kubernetes, Jenkins, GitHub Actions, Python, Scala, SQL, dbt, Spark (PySpark), Hive, Collibra, Great Expectations, IAM, Encryption, Role-Based Access Control, CloudWatch, Datadog, Splunk, DevSecOps

Aegis Software, Horsham, PA Jan 2020 – March 2023

Software Engineer (Data Engineer Focus)

Responsibilities:

- Translated complex data needs into robust **Spark (Scala) ETL pipelines**, ingesting **multi-terabyte datasets** in near **real-time**, enabling improved **analytics** performance.
- Developed **Snowflake** data warehouses, reducing **query durations by 50%** through partitioning, clustering, and advanced optimizations, while enforcing **strict governance** to meet enterprise **compliance standards** and ensure **reliable insights**.
- Automated **ingestion** from **SQL databases, Kafka, and REST APIs**, leveraging **Airflow** for near real-time data pipelines.
- Deployed **Docker** and **Kubernetes** microservices for **Java**, and **Python** applications, seamlessly integrating **AWS** platforms to support scalable, fault-tolerant solutions across multiple environments.
- Employed **AWS Lambda** for serverless computing, using **Terraform** to implement **infrastructure as code** and accelerate deployments, thereby significantly reducing manual overhead and configuration complexities.
- Upgraded **legacy SQL** systems to **Spark-centric frameworks**, boosting **scalability, concurrency**, and overall **performance** for massive data workloads through distributed processing.
- Constructed **event-driven pipelines** with **Kafka**, supporting asynchronous message flows, real-time data streaming, and rapid alerting to enhance operational responsiveness and reliability.

- Implemented **rigorous data validation, lineage tracking, and version control**, ensuring data integrity and audit readiness across evolving analytics demands and complex transformations.
- Developed and optimized data pipelines to ensure seamless data integration, enabling **Power BI dashboards** to deliver **real-time insights into key metrics and trends**, supporting **faster decision-making** for stakeholders across multiple business units.
- Implemented **secure authentication** with **OAuth2, encryption** at rest and in transit, plus **role-based access control** to effectively safeguard sensitive data and meet stringent compliance requirements.
- Collaborated with distributed teams, conducting thorough **testing** using **Postman for REST endpoints**, while facilitating **seamless integration** between front-end React components and back-end microservices across multiple environments.

Environment: Apache Spark (Scala), Snowflake, Kafka, RabbitMQ, Python, Java, JavaScript, Docker, Kubernetes, AWS (Lambda, Terraform), GCP, Apache Airflow, Power BI, OAuth2, Git, CI/CD Pipelines, Microservices Architecture.

Verisk Analytics, Jersey City, NJ June 2019 – Jan 2020

Associate Software Engineer

Responsibilities:

- Developed and maintained scalable **Java-based applications**, adhering to **industry best practices** and **design patterns**.
- Collaborated with cross-functional teams to clarify requirements, produce **technical specifications**, and deliver solutions aligned with **business objectives**.
- Wrote and refactored clean, efficient, and testable code, **proactively troubleshooting** existing codebases to enhance **performance** and **reliability**.
- Participated in **Agile ceremonies** (sprint planning, daily scrums, retrospectives) for **iterative development** and rapid **feedback loops**.
- Designed **test plans, test cases, and scripts**, performing both **manual** and **automated testing** across diverse environments.
- Developed **automated scripts** using **Java, Maven, TestNG, and Cucumber**, leveraging **Page Object Model (POM)** for structured test automation.
- Utilized **CI/CD pipelines** (Jenkins) for **nightly builds**, employing **Git** for version control and **faster feedback**.
- Conducted **RESTful API testing** with **Postman, Rest Assured**, and **SQL-based back-end** validations to **maintain data integrity**.
- Performed **Android/iOS app testing** on **Appium emulators** and **simulators**, ensuring a consistent user experience across mobile devices.

Education & Certifications

Bachelor of Science in Computer Science and Information Technology, Tribhuvan University (GPA: 3.7)

AWS Certified, **Data Engineer Associate**

Databricks Certified, **AWS Platform Architect**

Hashicorp Certified, **Terraform Associate**

Personal Interests

- **Tech Meetups & Conferences:** Stay current with emerging trends and network with industry experts by attending local and national events on cloud, DevOps, and big data.
- **Open-Source Contributions:** Contribute to GitHub repositories related to Python data libraries, enhancing code quality and documentation.
- **Reading & Research:** Passionate about reading technical blogs and books on distributed systems, AI ethics, and scalable architectures.
- **STEM Volunteering:** Mentor aspiring programmers at local code camps and hackathons, sharing knowledge on data engineering and DevOps practices.
- **Hiking & Travel:** Pursue outdoor adventures to recharge and gain fresh perspectives, often exploring national parks and scenic trails.

References Available Upon Request